

OBJECT-ORIENTED ANALYSIS & DESIGN

Object-Oriented Programming

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UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Object-Oriented Thinking
- 2 Design in the Software Process
- 3 Design for Quality Attributes
- 4 Objects



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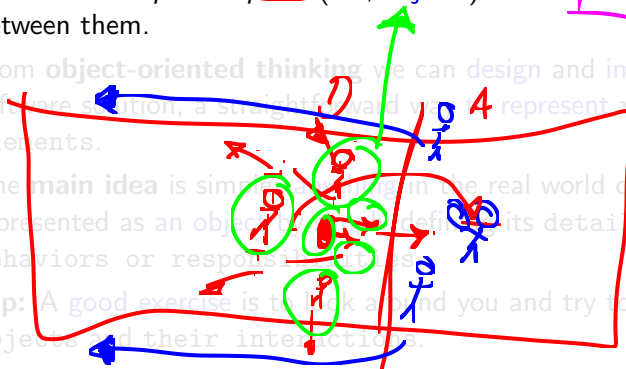


Object-Oriented

- **Object-oriented** is a **way of thinking** about problems. It is **not just** a way of programming.
- **Object-oriented thinking** involves **analyzing** a problem, **breaking it down** into *component parts* (i.e., **objects**) and the **interactions** between them.
- From **object-oriented thinking** we can **design** and **implement** a software solution, a straightforward way to **represent** real-world elements.
- The **main idea** is simple: **anything** in the real world can be represented as **an object** by simply defining its details and behaviors or responsibilities.
- **Tip:** A good exercise is to look around you and try to identify objects and their interactions.



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Jump

Run
Shoot

Algor. thms

KISS

functions

Color - overall
coins - accm
lives



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Software Implications

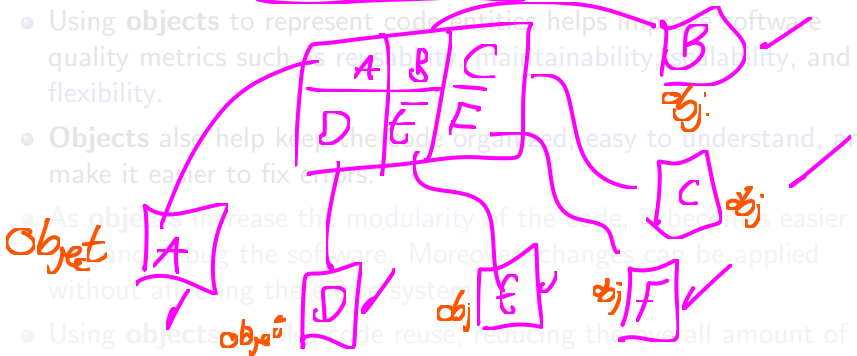
- Using **object-oriented thinking**, we can **model** a software system as a **collection of objects** that interact with each other. This approach applies a form of **divide and conquer** strategy.

- Using **objects** to represent code entities helps improve software quality metrics such as reusability, maintainability, scalability, and flexibility.

- Objects** also help keep the code organized, easy to understand, and make it easier to fix errors.

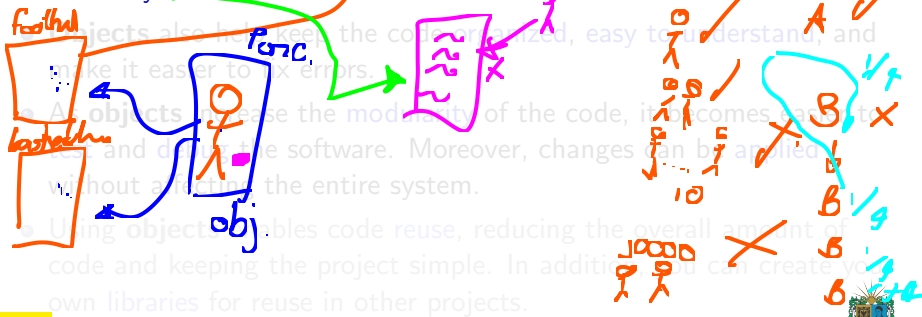
- As **objects** increase the modularity of the code, it becomes easier to maintain and debug the software. Moreover, changes can be applied without affecting the entire system.

- Using **objects** promotes code reuse, reducing the overall amount of code and keeping the project simple. In addition, you can create your own libraries for reuse in other projects.



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- As **objects** increase the **modularity** of the code, it becomes easier to **test** and **debug** the software. Moreover, changes can be **applied** without affecting the entire system.
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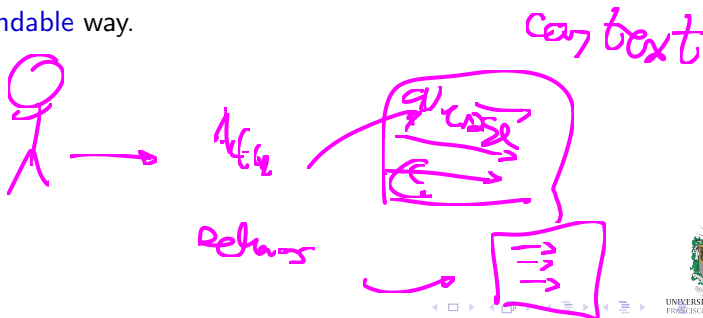
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What is abstraction?

- **Abstraction** is the process of **filtering out** the characteristics of an object that we are interested in, and **ignoring** the rest.
- **Abstraction** is a way to **simplify** the complexity of the real world by **focusing** on the relevant parts.
- **Abstraction** is a way to **represent** the essential features of an object, **hiding** the unnecessary details.
- **Abstraction** is a way to **model** the real world in a **simple** and **understandable** way.



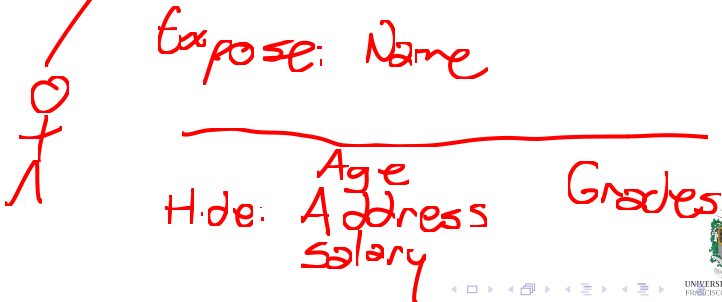
Abstraction Schemas

There are two types of **abstraction schemas**:

- **Data Abstraction:** This type of abstraction focuses on the data that an object contains. It is a way to **hide** the implementation details of an object and **expose** only the relevant data.

Behavior Abstraction: This type of abstraction focuses on the behavior of an object. It is a way to **hide** the implementation details of an object and **expose** only the relevant behavior.

common → details / attributes



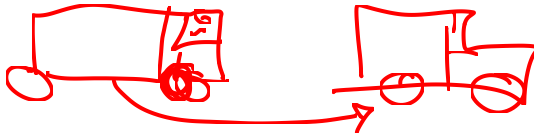
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algorithm
EU

USA



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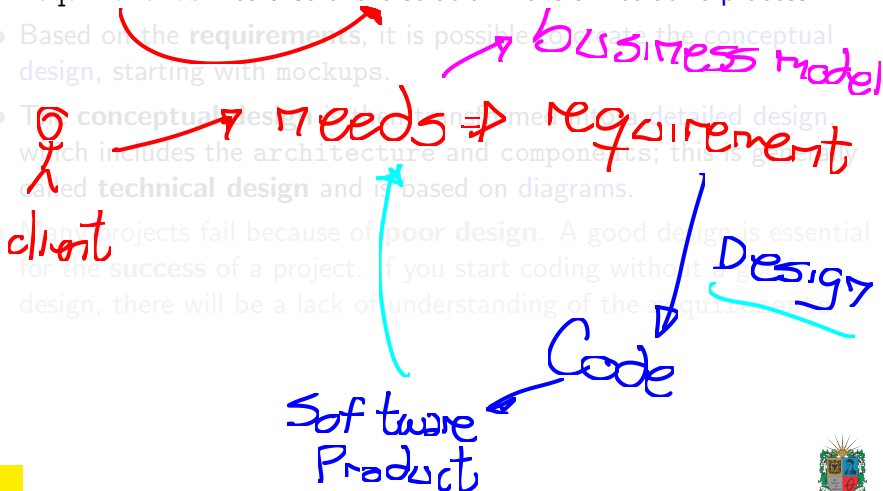
Software Design

- **Software Design** is the process of **transforming** a set of requirements into a **software solution**. It is an **iterative process**.

- Based on the **requirements**, it is possible to create the conceptual design, starting with mockups.

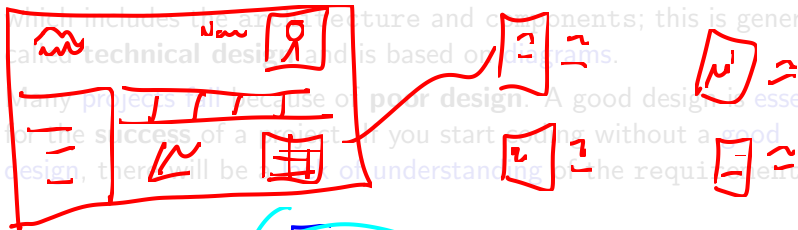
- The **conceptual design** is transformed into a detailed design, which includes the architecture and components; this is generally called **technical design** and is based on diagrams.

- Many projects fail because of **poor design**. A good design is essential for the success of a project. If you start coding without a good design, there will be a lack of understanding of the requirements.



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- The **conceptual design** is then transformed into a detailed design,



Figma / Balsamiq



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- Handwritten notes:*
 - "code & code design" (in red) with an arrow pointing from "technical design".
 - "Class Diagram" (in pink) with an arrow pointing from "diagrams".



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Requirements

Definition

Requirements are conditions or capabilities that must be implemented in a software product.

It is important to think like a software architect: consider both the structure and the behavior of the software.

- Requirements form the foundation of a software project. They define what the software should do and what the clients want (i.e., the scope).
- Elicit Requirements** is the process of gathering (i.e., asking the right questions) and documenting the needs of the clients.
- Functional Requirements** are the features that the software should have. They define what the software should do.
- Non-Functional Requirements** are the qualities that the software should exhibit. They define how the software should operate.



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Flowchart



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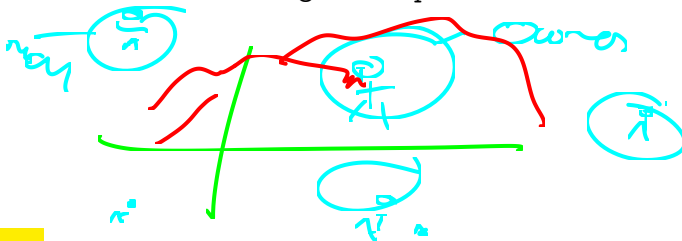
Conceptual Design

- Once the initial set of **requirements** are defined, the next step is to create a **conceptual design** of the software.
- **Conceptual Design** is a **high-level design** that defines the structure and behavior of the software. It is achieved by the **recognition** of the appropriate **components**, **connections**, and **responsibilities**.
- Conceptual Design is a visual representation of the software that helps communicate the design (layout, structure, and flow) to the stakeholders through mockups.

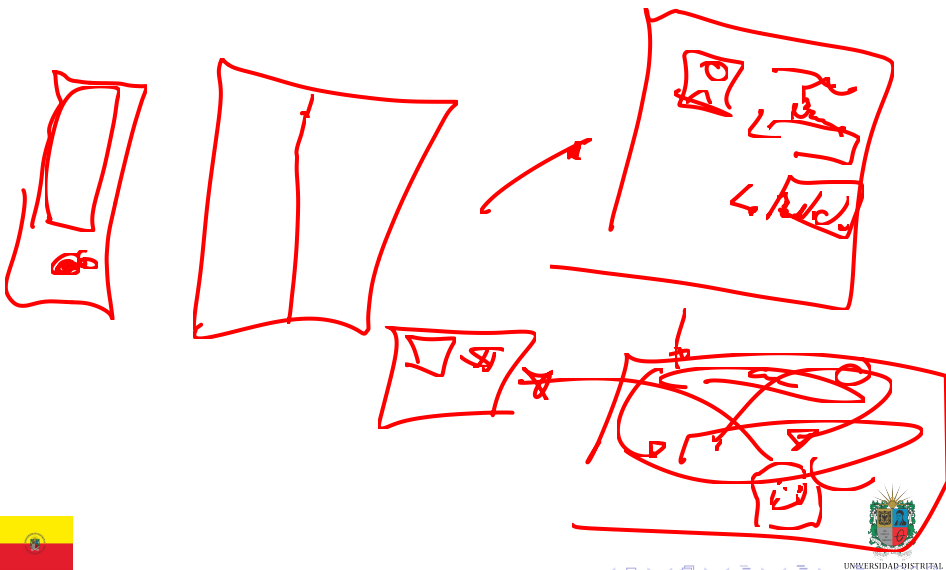


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Mockup Example: *Cell-Phone On-Line Store*



User Stories → requirements

- **User stories** are short, simple descriptions of a feature or function of a system.
- They are written from the perspective of the user and describe what the user wants to achieve.
- They are used to capture the requirements of a system in a simple and understandable way.

X no tech words



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☺ ■ expectations
 ↳ Acceptance
 Criteria



User Stories: Format Example

User Story

Title: Watch st. schedule	Priority:	Estimate:
User Story: As a [description of user], I want [functionality] so that [benefit].		
Acceptance Criteria: Given [how things begin] When [action taken] Then [outcome of taking action]		

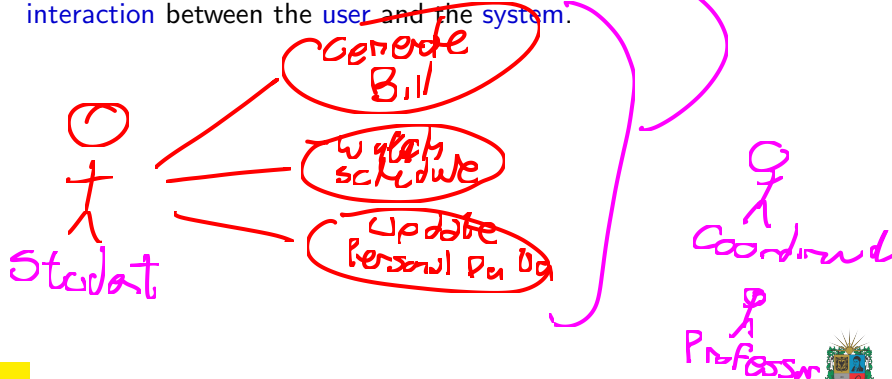
As 2 students,
I want to watch my schedule,
so that I can attend to my courses.

Given the enrolling start dates
when I do click in schedule button
Then my schedule should be
watching as a table

 ProductPlan


Use Cases

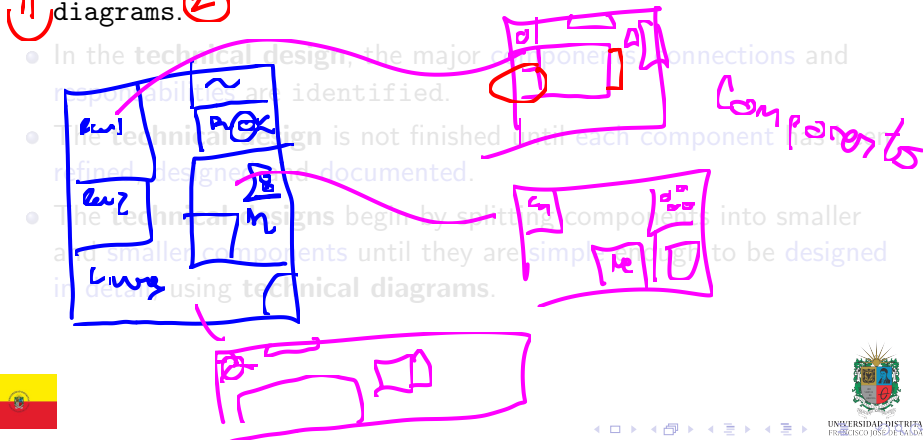
- **Use cases** are descriptions of how a system will be used by its users.
- They are used to capture the functional requirements of a system in a structured and detailed way.
- They are written from the perspective of the user and describe the interaction between the user and the system.



Technical Design

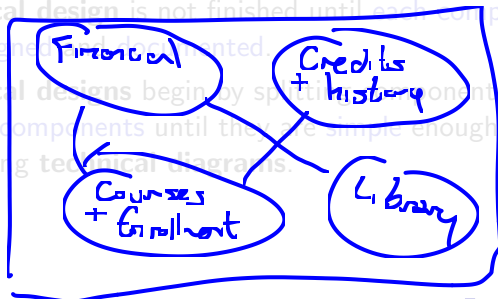
Full-stack

- Once the **(conceptual design)** is complete, the next step is to create a **technical design** of the software.
- Technical Design** is a **detailed design** that defines the architecture and components of the software. It is achieved by the creation of diagrams.



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- In the **technical design**, the major **components**, **connections** and **responsibilities** are identified.



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- In the **technical design**, the major **components**, **connections** and **responsibilities** are identified.
- The ~~technical design~~ is not finished until each component has been refined, designed and documented.
- The technical designs begin by splitting components into smaller and smaller components until they are simple enough to be designed in detail, using **technical diagrams**.



Compromise in Requirements and Design

- **Requirements** and **Design** are **interrelated**. **Requirements** are the **foundation** of the **design**. *needs*
- **Constant communication** and **feedback** is **key** to creating the **right solution** that **satisfies** the **client needs**.
- **Designs** will need to be reworked if **components**, **connections**, and the **responsibilities** of the **conceptual design** prove impossible to achieve in the **technical design**, or if they **fail to meet requirements**.
- **Larger systems** generally require **more time** to design, **more time** to implement, and **more time** to test.
- **Components** at this stage may be *refined enough* to become **collections** of functions, classes, or other components. These pieces become a more **manageable problem** that **developers** can individually **implement**.



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Trade-Offs

- Trade-offs are inevitable in software design. Quality attributes are often competing and contradictory.
- Trade-offs are necessary to balance the competing quality attributes of a system.
- Trade-offs are made by weighing the importance of each quality attribute and deciding which attribute is more important.
- Trade-offs are made by compromising on less important quality attributes in order to improve the more important attributes.
- Trade-offs are necessary to create a system that satisfies the needs of the users and stakeholders.

Security → speed
+ auth — performance
slow



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- **Trade-offs** are **inevitable** in software design. **Quality attributes** are often **competing** and **contradictory**.
- **Trade-offs** are *necessary* to *balance* the competing **quality attributes** of a **system**.
- **Trade-offs** are *made* by **weighing** the **importance** of each **quality attribute** and *deciding* which **attributes** are **most important**.
- **Trade-offs** are *made* by **compromising** on **less important quality attributes** in order to improve the **more important attributes**.
- Trade-offs are **necessary** to *create* a **system** that *satisfies* the **needs** of the **users** and **stakeholders**.

sec. 0.6

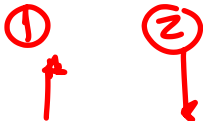
Fast 0.9

Frondly 0.2



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Context and Consequences

- **Context** is the environment in which a system will be used. It includes the users, the stakeholders, and the constraints of the system.
- **Context** provides important information when deciding on the balance of qualities in design.
- **Consequences** are the results of the decisions that are made during the design of a system. They include the trade-offs that are made and the impact that they have on the system. *Handwritten: DV....*
- A **good practice** is to seek other perspectives on technical designs. This can be done by asking other developers for their opinion, or by having a design review session. *Handwritten: 3rd/1st Day*
- Another **good practice** is to perform prototyping and simulation to test the design before implementing it.



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→ experience



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code
(Figma)
Balsamiq ← *mockups*



Satisfying Qualities

- **Quality attributes** are the characteristics of a **system** that *determine* its **quality**. They are the features that *define* how well a **system** *satisfies* the **needs** of its users.
- **Quality attributes** are **important** because they *determine* how well a system will **satisfy** the **needs** of its users.
- **Quality attributes** have a strong *relationship* with the **non-functional requirements** to satisfy aspects as **performance**, **resource usage** and **efficiency**.
- Other **qualities** that software often satisfies in non-functional requirements include **reusability**, **flexibility**, and **maintainability**. This helps inform *how well* the code of software can evolve and allow for *future changes*.

level



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- Handwritten red note:* + $\Rightarrow$ global quality



Satisfying Qualities

- **Quality attributes** are the **characteristics** of a **system** that *determine* its **quality**. They are the **features** that *define* how well a **system** *satisfies* the **needs** of its users.
- **Quality attributes** are **important** because they *determine* how well a **system** will **satisfy** the **needs** of its users.
- **Quality attributes** have a **strong relationship** with the non-functional requirements to satisfy aspects as **performance**, **resource usage** and **efficiency**.

Req. Func.

- slow schedule
- enroll course
- pay debt

Req. non-Func.

- Resp < 300 ms
- Support > 5000 users
- Enqueue no loss request



Satisfying Qualities

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- **Quality attributes** are **important** because they *determine* how well a **system** will **satisfy** the **needs** of its users.
- **Quality attributes** have a strong *relationship* with the non-functional requirements to satisfy aspects as **performance**, **resource usage** and **efficiency**.
- Other **qualities** that **software** often satisfies in non-functional requirements include **reusability**, **flexibility**, and **maintainability**. This helps inform *how well* the code of software can **evolve** and allow for *future changes*.



Outline

- 1 Object-Oriented Thinking
- 2 Design in the Software Process
- 3 Design for Quality Attributes
- 4 **Objects**



Basics of Object-Oriented Design I

- **Object-oriented** has become one of the most traditional and popular **paradigms** in software development.
- It is based on the concept of **objects**, which can contain data, in the form of **fields** (often known as **attributes** or **properties**), and code, in the form of **procedures** (often known as **methods**).

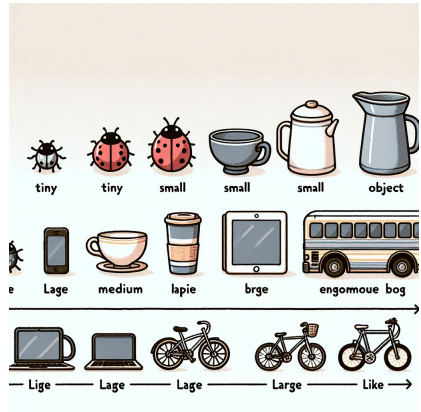


Figure: Prompt: Draw several objects sorted by size.



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behaviors

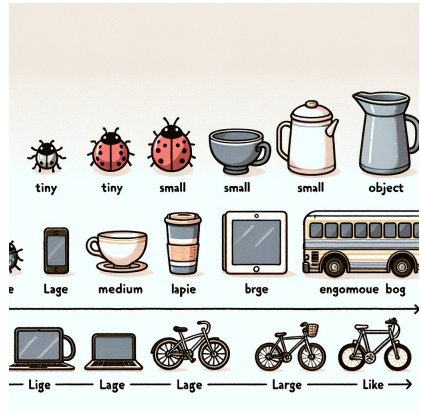


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Basics of Object-Oriented Design II

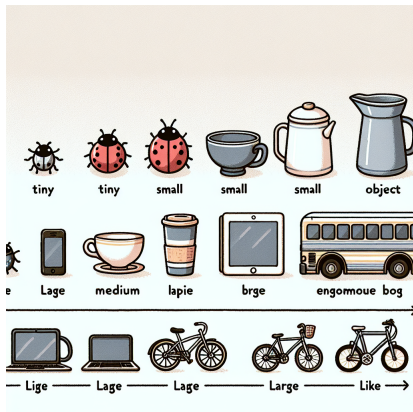
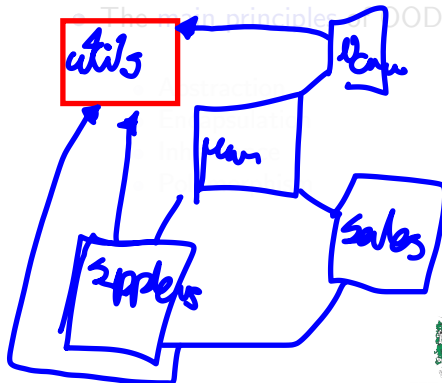


Figure: Prompt: Draw several objects sorted by size.

- The idea is to **design** a **system modularly**, making it easier to maintain and understand. Another key idea is to emphasize **code reuse**.



Basics of Object-Oriented Design II

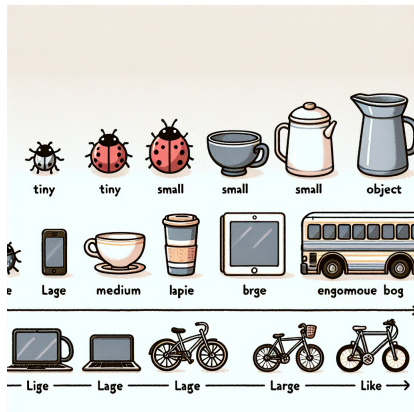


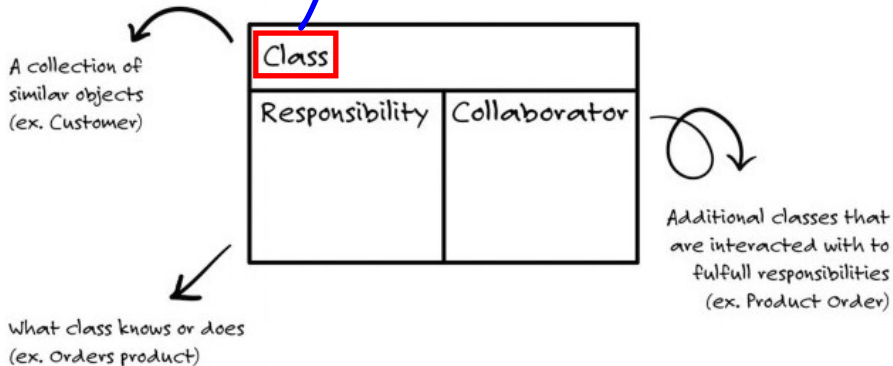
Figure: Prompt: Draw several objects sorted by size.

- The idea is to **design** a **system modularly**, making it easier to maintain and understand. Another key idea is to emphasize **code reuse**.
- The **main principles** of OOD are:
 - Abstraction
 - Encapsulation
 - Inheritance
 - Polymorphism



CRC Cards

class solution



Prototyping and Simulation

- **CRC cards** are useful tools, especially for prototyping and simulation during conceptual design.
- **CRC cards** are excellent tools to bring to software development team meetings. All the cards can be placed on the table to facilitate discussion or simulate how these classes work together with other classes to achieve their responsibilities.

Manual



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/object-oriented-programming>

